

Exhibiting Evidence in Remote Courtrooms: Design and Usability Study

Anurata Prabha Hridi*

Khawar Latif Khan*

aphridi@ncsu.edu

kkhan3@ncsu.edu

North Carolina State University

Raleigh, North Carolina, USA

Saurabh Nanda

North Carolina State University

Raleigh, North Carolina, USA

snanda2@ncsu.edu

Aditya Ravikant Jadhav

North Carolina State University

Raleigh, North Carolina, USA

ajadhav3@ncsu.edu

Leanne Serrao

North Carolina State University

Raleigh, North Carolina, USA

lserrao@ncsu.edu

Hossein Saedi

North Carolina State University

Raleigh, North Carolina, USA

hsaedi@ncsu.edu

Benjamin Watson

North Carolina State University

Raleigh, North Carolina, USA

bwatson@ncsu.edu

ABSTRACT

Remote court proceedings, an intensely collaborative process, are a reality in the pandemic but struggle to support evidence review. Remote proceedings are likely to continue after the pandemic, making it imperative to strengthen evidence viewing and analysis for jurors. Our project addressed this problem by prototyping an assistive application that offered multiple points of view and evaluating it in a small user study. Five participants used the application to review case-specific exhibits in a hypothetical court session while thinking aloud. Participants were satisfied with the application, with almost no facilitation. We also identified the need to make a display of the setting in which the evidence was found more interactive and navigable.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative interaction.**

KEYWORDS

User experience, remote courtroom, jury, evidence.

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1 INTRODUCTION AND BACKGROUND

Remote court proceedings were uncommon before the pandemic, but have since become quite frequent [19] and are likely to continue

[23], because they improve efficiency, cost, and access [10, 25]. The court needs specific, purpose-built solutions for the long term.

Perhaps surprisingly, research on augmenting legal practice with technology predates the pandemic by many years [5, 7, 22]. Virtual reality (VR) and 3D graphics have been extensively used to teach criminal law [13] and crime scene investigation [12], train people for court appearances [17], create a safe virtual room for the exchange of important legal information [6], and enable preparation and presentation for the court [2]. Nordin & Zekan did impressive work on the use of VR in court but did not envision the court itself in VR [16]. Even before the pandemic, researchers noted that remote attendance could reduce cost, improve access and also reduce bias [14, 20, 21].

US courts struggled during the pandemic [1], with some states cancelling in-person trials [8]. Remote proceedings surged [10], but the virtual court experience was not as engaging, and interactive [3], a major shortcoming for counsel seeking to persuade judges and juries.

Digital examination of evidence was also explored well before the pandemic, with research showing that it could be helpful [11], particularly when displaying remote scenes and reenactments. The pandemic often required all evidence to be demonstrated digitally, driving new calls for research [18, 24] to restore information and constraints often lost in the digital transformation to remote interaction. Bender [4] makes the obvious suggestion of using video to present evidence, but does not analyze any of its shortcomings or advantages: the inability to change view, feel the heft and texture of an object; or the ability to present evidentiary context, including date and location of discovery, provenance and composition, and annotation of relevant features.

We sought to develop a solution allowing jurors to examine evidence digitally and remotely. To mitigate the disadvantages of such a solution, it would allow viewing evidence from all angles and textually describing its weight and surface characteristics. To maximize the advantages, it would support the display of evidence in the scene in which it was discovered, as well as a textual description of provenance and important features. Section 2 briefly describes our design process, including the illustration of the prototype. We discuss our think-aloud user study in Section 3 before concluding in Section 4.

*Both authors contributed equally to this research.

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Figure 1: Phases during trial hearing

2 DESIGN PROCESS

This work was performed for a project in a user experience course at NC State University. We adopted the five-step sprint presented by Knapp et al. [9]. The design process and decisions were guided by discussions with legal experts and analysis of prior research that helped us understand user problems and design constraints. Additionally, an inspection of the existing video conferencing tools, e.g., Zoom, Skype, and Microsoft Teams, among others, highlighted their drawbacks.

We then created an experience map describing how jury members interact with evidence. In particular, we noted that jurors study evidence in two phases: in the hearing phase, they examine evidence in the order it is brought to them by counsel, while in the deliberation phase, they examine previously exhibited evidence as needed, in the order they like. We next identified those elements of the experience map that were likely the most challenging to perform remotely and diverged by generating a surplus of potential solutions addressing those elements. We converged by elaborating

on the most promising of these solutions in storyboards and choosing among them. We prototyped our chosen solution by identifying an evidentiary scenario, matching content and the tools we would use, and then building the prototype.

Finally, we performed a lightweight usability test to reveal remaining issues with our prototype, using a think-aloud protocol.

2.1 Prototype and User Journey

The interactive prototype created in Figma can be accessed at <https://bit.ly/ev-dash>. Figures 1 and 2 illustrate the workflow during the hearing and deliberation phase, respectively. Using the prototype's interface panel, jurors can access all the pieces of evidence presented in the trial, either individually or collectively, depending on the trial phase.

Figure 3 shows the key features of the prototype. The control panel at the bottom allows jurors to study the evidence in detail by changing viewing angles, zooming in and out, and placing the object at the scene of discovery (e.g. "the crime scene"). The column

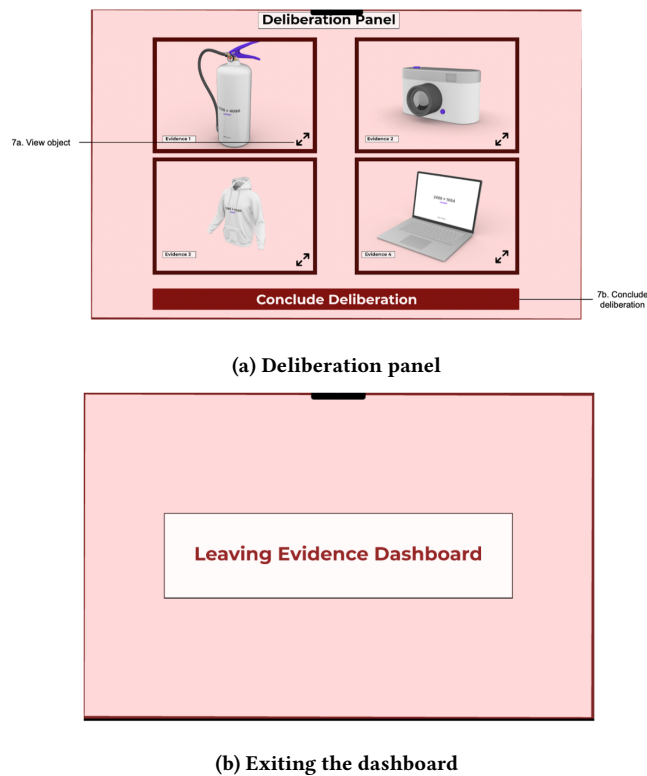


Figure 2: Phases during trial deliberation

on the right provides other relevant information about the object, such as the date and location of discovery, provenance, etc.

Note that although our prototype extensively uses the color red, it does not rely on red/green contrast, which is challenging to perceive for those with the most common form of color vision deficiency.

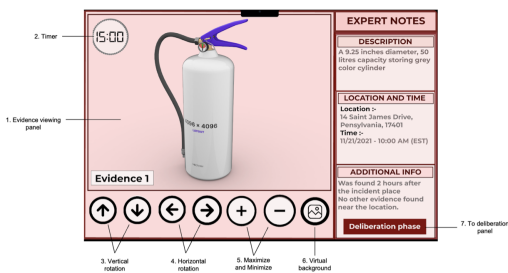


Figure 3: Prototype functionalities

During the hearing phase, jurors log in and see the first exhibited evidence on the dashboard. As each piece of evidence is introduced, a timer indicates the time left to view it. Jurors can hover over the evidence to view notes and expert annotations or use the side panel to zoom the view in, out, and rotate. With a button press, jurors can view the evidence at the scene of the discovery, gaining a better understanding of the context in which the object was found.

If time expires, jurors may request an extension. Evidence may take different forms, including written documents, videos, audio recordings, or physical objects such as weapons, garments, etc.

During the deliberation phase, the prototype dashboard displays all the evidence introduced during the trial. Evidence may be viewed in the order desired, with each piece's details viewed as it was during the hearing phase. Combined with the display of the scene of the discovery, this gives jurors a holistic view of the case.

3 USABILITY STUDY

To reveal the remaining issues with our prototyped solution for remote evidence examination, we followed the think-aloud methodology described by Nielsen [15]. We recruited five student participants (two females and three males) from various fields of study at NC State University with announcements on social media, requiring only that they speak English. After participants gave informed consent, we scheduled a 20-minute session with each participant.

3.1 Tasks

We asked participants to perform three tasks using our prototype in the trial and deliberation phases. We encouraged participants to think aloud as they did so, which helped us understand the problems and successes they encountered. We gave our participants the following tasks, each difficult to perform with a standard video conferencing system:

- *Task 1* was to take a closer look at a piece of evidence from different angles and perspectives.
- *Task 2* was to change the background to view the evidence in the scene of its discovery.
- *Task 3* was to deliberate by viewing pieces of evidence in the order they desired.

3.2 Issues and Successes

All users were able to perform these tasks successfully, with almost no facilitation. The thorough examination of *Task 1* made it the longest to perform, while the rich context of *Task 2* satisfied participants. We identified the following areas of improvement:

- *Task 1*: When viewing individual pieces of evidence
 - When attempting to rotate the view of the object, almost all participants tried to click on the image of the object and rotate it using their mouse before attempting to use the control panel.
 - Our prototype only zoomed into the center of the current view. Some participants attempted to zoom into arbitrary features on the evidence.
- *Task 2*: When viewing the evidence in the context of its discovery scene
 - Some participants did not understand the icon on the scene of the discovery button.
 - Some participants tried to rotate the piece of evidence in this context, which the prototype did not allow.
- *Task 3*: When deliberating
 - Almost all participants tried to view the individual evidence by clicking on it rather than clicking on the full-screen button.

4 LIMITATIONS AND FUTURE WORK

Our prototype for remote examination of evidence showed promise. Participants were consistently able to perform three challenging tasks with video conferencing tools: examining evidence from various viewpoints, at the scene of the discovery, and together achieving a holistic understanding of the case. Nevertheless, this was only a preliminary design and evaluation; a study with a higher fidelity prototype, a richer task context, and more participants would add certainty and generalizability. Our evaluation also revealed issues that should be addressed, particularly with view control. We believe most of these issues could be addressed by using a true 3D model of the evidence, permitting arbitrary rotations and zooms, and intuitive direct manipulation of the evidence itself, including touch interaction.

Longer term, our work revealed several shortcomings of all forms of remote evidence examination, including our prototype. During the hearing phase, counsel will point to evidence to illustrate their arguments immediately in front of the jury. This combination can be particularly compelling and should be delivered as faithfully as possible to remote juries. The weight and feel of evidence in jury hands can be very informative and currently cannot be communicated intuitively. Finally, the remote examination also has strengths that should be emphasized further, including extreme close-ups, viewing in different lighting, times of day and weather conditions, and 3D reenactments.

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REFERENCES

- [1] Deniz Ariturk, William E Crozier, and Brandon L Garrett. 2020. Virtual criminal courts. *U. Chi. L. Rev. Online* (2020), 57.
- [2] Jeremy N Bailenson, Jim Blascovich, Andrew C Beall, and Beth Noveck. 2006. Courtroom applications of virtual environments, immersive virtual environments, and collaborative virtual environments. *Law & Policy* 28, 2 (2006), 249–270.
- [3] Susan A Bandes and Neal Feigenson. 2020. Virtual trials: Necessity, invention, and the evolution of the courtroom. *Buff. L. Rev.* 68 (2020), 1275.
- [4] Matthew Bender. 2021. Unmuted: Solutions to Safeguard Constitutional Rights in Virtual Courtrooms and How Technology Can Expand Access to Quality Counsel and Transparency in the Criminal Justice System. *Vill. L. Rev.* 66 (2021), 1.
- [5] Chantell Bergquist. 2021. Virtual Hearings and Blockchain Technology Solutions in Criminal Law. *Mitchell Hamline Law Review* 47, 3 (2021), 10.
- [6] Sabine Cikir, Sabina Jeschke, Fritz Lehmann-Grube, and Jan Sablatnig. 2008. Using a room metaphor for e-forensic working environments. In *Proceedings of the 1st international conference on Forensic applications and techniques in telecommunications, information, and multimedia and workshop*. Citeseer, 1–5.
- [7] Ken Fowle and Damian Schofield. 2013. Technology corner visualising forensic data: evidence guidelines (Part 1) *Journal of Digital Forensics. Security and Law* 8, 2 (2013), 93–114.
- [8] Sarah Jarvis. 2020. Coronavirus: The Latest Court Closures And Restrictions. *law360*. <https://www.law360.com/articles/1252836/coronavirus-the-latest-court-closures-and-restrictions>
- [9] Jake Knapp, John Zeratsky, and Braden Kowitz. 2016. *Sprint: How to solve big problems and test new ideas in just five days*. Simon and Schuster.
- [10] Amy-May Leach, Lyndsay R Woolridge, Brian L Cutler, Jeffrey S Neuschatz, and Baylee D Jenkins. 2021. COVID-19 and the courtroom: how social and cognitive psychological processes might affect trials during a pandemic. *Psychology, Crime & Law* (2021), 1–32.
- [11] Fredric I Lederer. 1998. The Road to the Virtual Courtroom—a Consideration of Today's—And Tomorrow's—High-Technology Courtrooms. *SCL Rev.* 50 (1998), 799.
- [12] Richard Mayne and Helen Green. 2020. Virtual reality for teaching and learning in crime scene investigation. *Science & justice* 60, 5 (2020), 466–472.
- [13] Markos Mentzelopoulos, James Parrish, Paresh Kathrani, and Daphne Economou. 2016. REVRLaw: An immersive way for teaching criminal law using virtual reality. In *International Conference on Immersive Learning*. Springer, 73–84.
- [14] Linda Mulcahy, Emma Rowden, and Wend Teeder. 2020. Testing the case for a virtual courtroom with a physical jury hub: Second evaluation of a virtual trial pilot study conducted by JUSTICE. Available at SSRN 3876209 (2020).
- [15] Jakob Nielsen. 2000. Why You Only Need to Test with 5 Users. Nielsen Norman Group. <https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/>
- [16] Carolina Nordin and Adna Zekan. 2021. Taking Virtual Reality into the Courtroom: An exploratory study of how to implement Virtual Reality within Swedish court. (2021).
- [17] Helga Margrét Ólafsdóttir, Edit Ómarsdóttir, Hafþís Sæland, and Hannes Högni Vilhjálmsson. 2019. Work in Progress Report: Virtual Courtroom to Prepare Victims of Sex Crimes for Court Proceedings. In *Proceedings of the 19th ACM International Conference on Intelligent Virtual Agents*. 170–172.
- [18] Meredith Rossner, David Tait, and Martha McCurdy. 2021. Justice reimaged: challenges and opportunities with implementing virtual courts. *Current Issues in Criminal Justice* 33, 1 (2021), 94–110.
- [19] Steve Rubley. 2021. Forum Infographic: The COVID-19 pandemic and the courts — Aggravation or opportunity? Thomson Reuters Special Services. <https://www.thomsonreuters.com/en-us/posts/legal/forum-infographic-fall21-pandemic-and-the-courts/>
- [20] Natalie Salmanowitz. 2016. Unconventional methods for a traditional setting: The use of virtual reality to reduce implicit racial bias in the courtroom. *UNHL Rev.* 15 (2016), 117.
- [21] Natalie Salmanowitz. 2018. The impact of virtual reality on implicit racial bias and mock legal decisions. *Journal of Law and the Biosciences* 5, 1 (2018), 174–203.
- [22] Damian Schofield and Ken Fowle. 2013. Technology corner: visualising forensic data: evidence guidelines (Part 2). (2013).
- [23] Jacqueline Thomsen. 2021. Virtual Court Hearings Are Here to Stay Post-Pandemic, Survey Finds. Legal Technology. <https://www.law.com/nationalallawjournal/2021/08/18/virtual-court-hearings-are-here-to-stay-post-pandemic-survey-finds/?slreturn=20220704023437>
- [24] Elizabeth G Thornburg. 2020. Observing Online Courts: Lessons from the Pandemic. *Family Law Quarterly* 54, 3 (2020), 181–244.
- [25] Jenia I Turner. 2020. Remote criminal justice. *Tex. Tech L. Rev.* 53 (2020), 197.